

Signal efficiencies in the limit $p_T \rightarrow \infty$

How the fit of $\varepsilon_\infty + c/p_T^2$ to the muons above 10 GeV/c works

Muon Collider BIB study · 30 September 2026 · numbers from run 2026-09-30_145059_pdf (the current cuts and resolutions)

The signal hit efficiency and the track-finding efficiency quoted in the results (the summary table and the highlights PDF) are values in the limit $p_T \rightarrow \infty$. Every trial – a signal hit, or a muon – with generated p_T above 10 GeV/c becomes one data point, at $x = 1/p_T^2$, with $y = 1$ if it passes and $y = 0$ if it fails. A straight line $y = \varepsilon_\infty + c x$ is fitted through all these points by least squares, and ε_∞ is its intercept at $x = 0$, i.e. at $p_T \rightarrow \infty$. With the current cuts the line comes out flat within its errors, so ε_∞ is essentially the efficiency of the highest- p_T muons, with an error that does not assume the flatness.

1 The method, step by step

- 1. The trials.** For the signal hit efficiency of a subsystem, each trial is one of the muon's own hits in that subsystem (hits of its secondaries are left out, `muon_hits_only = true`). Its outcome is whether it passes all three cuts of that subsystem – time, z_0 and p_T – and its p_T is the generated transverse momentum of the muon that made it, $p_T = \sqrt{p_x^2 + p_y^2}$ from the Monte Carlo truth (`part_px`, `part_py`; one muon per event). The totals VXD, IT + OT and ALL pool the hits of their subsystems. For the track-finding efficiency, each trial is one muon, and its outcome is *found*: at least 5 of its hits survive the cuts outside the vertex detector.
- 2. The fit range.** Only trials with $p_T > p_{\min}$ enter, where $p_{\min}$ is the larger of 10 GeV/c and twice the largest p_T cut among the subsystems involved. The largest p_T cut is now 2 GeV/c (OT only), so $p_{\min} = 10$ GeV/c everywhere.
- 3. The fit.** With $x_i = 1/p_{T,i}^2$ and $y_i \in \{0, 1\}$, one term per trial in the fit range,

$$S(\varepsilon_\infty, c) = \sum_i (y_i - \varepsilon_\infty - c x_i)^2 \quad (1)$$

is minimized. The minimum follows in closed form from the normal equations,

$$\begin{pmatrix} \varepsilon_\infty \\ c \end{pmatrix} = (X^T X)^{-1} X^T y, \quad X = \begin{pmatrix} 1 & x_1 \\ \vdots & \vdots \\ 1 & x_n \end{pmatrix}, \quad (2)$$

so there are no bins, no starting values and no choice other than $p_{\min}$. Since each y_i is 0 or 1, the line fits the mean efficiency as a function of x . A binomial maximum-likelihood fit of the same line gives the same ε_∞ to within 0.01 % (checked for the tracks, the VXD barrel and ALL).

- 4. The uncertainty.** The covariance matrix of (ε_∞, c) is the “sandwich” estimator

$$V = (X^T X)^{-1} \left(\sum_g s_g s_g^T \right) (X^T X)^{-1}, \quad s_g = \sum_{i \in g} r_i \begin{pmatrix} 1 \\ x_i \end{pmatrix}, \quad r_i = y_i - \varepsilon_\infty - c x_i, \quad (3)$$

and $\sigma(\varepsilon_\infty) = \sqrt{V_{11}}$. It is built from the actual scatter of the residuals, so it does not assume a binomial variance. For the hit efficiencies each group g is one muon: the residuals of its hits are summed before squaring, because the hits of one muon are not independent. For track-finding each muon is its own group. In practice the grouping changes little (VXD barrel: 0.093 % $\rightarrow$ 0.094 %; IT endcap: 0.115 % $\rightarrow$ 0.122 %). A result above 100 % would be set to 100 %.

In the code: `efficiency_at_infinite_pt` and `pt_inf_fit_min` in `bib_common.py`, called by `apply_cuts.py` (each subsystem, the VXD, IT + OT and ALL totals, and the per-cut tables) and by `track_efficiency.py`.

2 Why $1/p_T^2$ and not $1/p_T$

Expand the efficiency in the signed curvature $\kappa = q/p_T$ around $\kappa = 0$: $\varepsilon(\kappa) = \varepsilon_\infty + a\kappa + b\kappa^2 + O(\kappa^3)$. The sample holds as many μ^+ as μ^- (50 145 and 49 855), so at a given p_T what is measured is the average over the two charges,

$$\frac{1}{2}[\varepsilon(\kappa) + \varepsilon(-\kappa)] = \varepsilon_\infty + b\kappa^2 + O(\kappa^4) = \varepsilon_\infty + \frac{c}{p_T^2} + \dots \quad (4)$$

The linear term cancels exactly, even if the detector responds differently to the two charges. Physically, the leading effects of a finite momentum all go as $1/p^2$:

- multiple scattering deflects the muon either way by an rms angle $\propto 1/p$, so the losses it causes go as its variance, $\propto 1/p^2$;
- the velocity, $\beta \simeq 1 - m_\mu^2/2p^2$, enters the time of flight;
- the curvature changes the path length (an arc exceeds its chord by $\simeq s^3/24R^2$) and the straight-line extrapolation to z_0 , both by amounts $\propto 1/R^2 \propto 1/p_T^2$.

The gun's θ distribution does not depend on p_T , so averaging over θ turns $1/p^2$ into c/p_T^2 with an effective c .

3 Why the fit starts at 10 GeV/c

The expansion only holds well above the turn-on of the p_T cut, where the efficiency changes steeply. For OT barrel hits it is 3.6 % at 1.5–2 GeV/c, 85.4 % at 2–2.5 GeV/c and 99.3 % at 2.5–3 GeV/c before it flattens out; the track-finding efficiency has the same turn-on (Fig. 1a). Hence the rule: at least twice the largest p_T cut, and never below 10 GeV/c.

4 Why the extrapolation is short

The muon gun is flat in $1/p_T$, from $1/1.5$ (GeV/c) $^{-1}$ down to almost 0: the highest p_T in the sample is 120 TeV/c. So 15 % of the muons are above 10 GeV/c (14 974 of 100 000), and of those one third are above 30 GeV/c and one tenth above 100 GeV/c. In $x = 1/p_T^2$ the fit covers $0 < x < 0.01$ (GeV/c) $^{-2}$, with a density $\propto x^{-1/2}$: the points crowd towards $x = 0$ (Fig. 1b). The intercept is pinned by real data at very high p_T ; the slope term only protects it from a bias.

5 Example: the track-finding efficiency

The fit gives

$$\varepsilon_\infty = (99.08 \pm 0.11) \%, \quad c = (0.26 \pm 0.23) (\text{GeV}/c)^2,$$

from 14 974 muons (Table 1, Fig. 1).

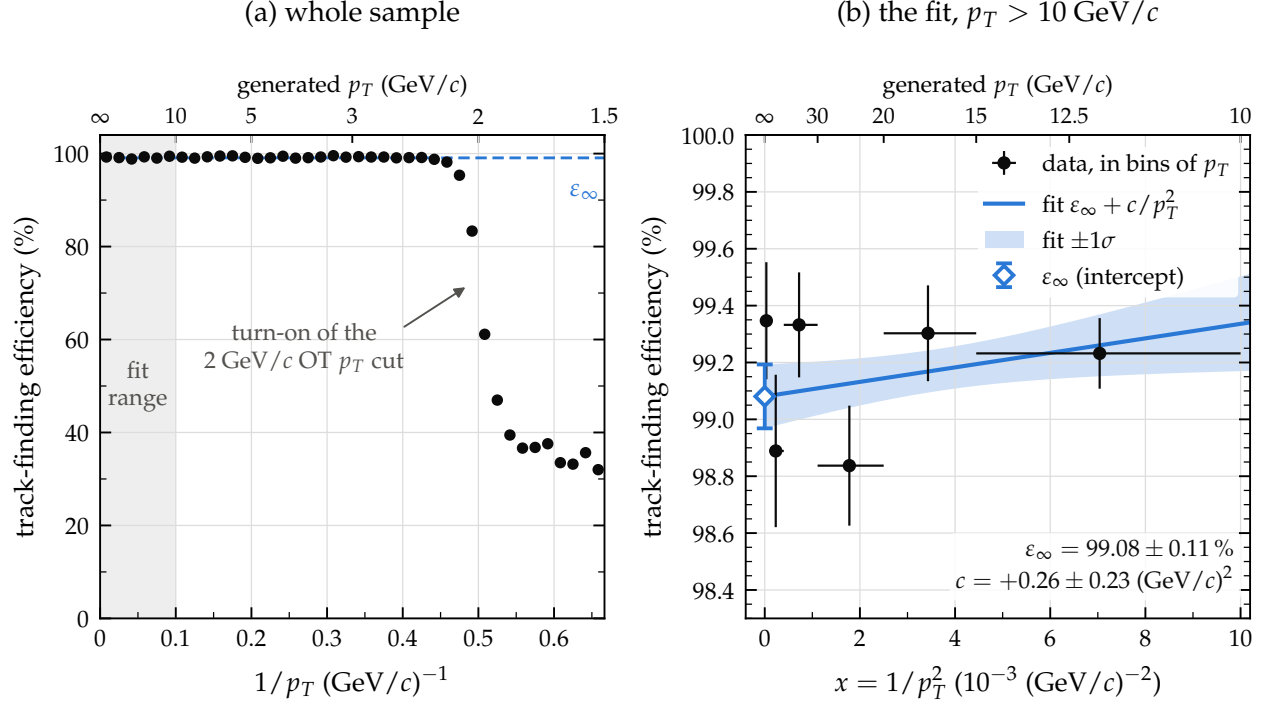


Figure 1: Track-finding efficiency (found: at least 5 surviving hits outside the vertex detector). (a) The whole sample against $1/p_T$, in which the gun is flat, so every bin holds about 2 500 muons. The efficiency turns on at the 2 GeV/c p_T cut of the OT and is flat above it; shaded: the fit range, $p_T > 10 \text{ GeV}/c$; dashed: ε_∞ . (b) The fit range against $x = 1/p_T^2$ (top axis: p_T). The fit uses the individual muons; the points, in bins of p_T (10–15, 15–20, 20–30, 30–50, 50–100 and above 100 GeV/c, drawn at their mean x , horizontal bars: bin extent), are for display. Line and band: the fitted $\varepsilon_\infty + c/p_T^2$ and its $\pm 1\sigma$ band; diamond: ε_∞ .

- The slope is compatible with zero: above 10 GeV/c the efficiency is flat within its errors. ε_∞ is the plain average above 10 GeV/c (99.17 %) minus $c\langle x \rangle = 0.08 \%$.
- Its error is 1.5 times that of the plain average $\bar{\varepsilon}$. For a straight-line fit

$$\sigma(\varepsilon_\infty) = \sigma(\bar{\varepsilon}) \sqrt{1 + \langle x \rangle^2 / \text{Var}(x)},$$

and here $\langle x \rangle = 0.0033$ and $\text{rms}(x) = 0.0030 (\text{GeV}/c)^{-2}$. This is the price of fitting the slope instead of assuming it is zero.

- The average over the whole sample, 83.7 %, mostly measures how many muons lie below the 2 GeV/c turn-on. That is why the limit is quoted.
- Moving the start of the fit changes ε_∞ by at most 0.16 %, within its errors: starting at 5, 7, 10, 15, 20, 40 GeV/c gives $\varepsilon_\infty = 99.12, 99.14, 99.08, 99.10, 99.24, 99.21 \%$, with errors from 0.08 to 0.20 % (the fits share muons, so these values are not independent).

Table 1: Track-finding efficiency in bins of generated p_T (display only: the fit uses the individual muons above 10 GeV/c). Last column: the fitted line at each bin's mean $1/p_T^2$.

generated p_T (GeV/c)	muons	found	efficiency (%)	fit line (%)
1.5–2	24 878	9 732	39.12 ± 0.31	–
2–2.5	15 051	14 391	95.61 ± 0.17	–
2.5–5	30 118	29 884	99.22 ± 0.05	–
5–10	14 979	14 873	99.29 ± 0.07	–
10–15	4 948	4 910	99.23 ± 0.12	99.26
15–20	2 438	2 421	99.30 ± 0.17	99.17
20–30	2 580	2 550	98.84 ± 0.21	99.13
30–50	1 947	1 934	99.33 ± 0.18	99.10
50–100	1 530	1 513	98.89 ± 0.27	99.09
above 100	1 531	1 521	99.35 ± 0.21	99.08
all	100 000	83 729	83.73	

6 The signal hit efficiencies

The same fit is made for each subsystem and for the totals (Table 2, Fig. 2). In every subsystem the slope is compatible with zero within 2σ . Where no p_T cut acts, the hit efficiency is flat far below 10 GeV/c as well: in the VXD barrel it is $97.75 \pm 0.05\%$ at 1.5–2 GeV/c and $97.93 \pm 0.11\%$ at 20–50 GeV/c. That loss is mostly the Gaussian tail of the 0.03 ns time smearing outside the ± 0.07 ns cut: $\pm 2.3\sigma$ alone gives 98.0%, and it does not depend on p_T . In the OT the whole-sample averages (71–72%) are again dominated by the turn-on of the 2 GeV/c cut, while the limits are 99.93% (barrel) and 99.56% (endcap).

Table 2: Signal hit efficiency for $p_T \rightarrow \infty$: hits and muons in the fit ($p_T > 10$ GeV/c), the fit results, and, for comparison, the plain averages above 10 GeV/c and over the whole sample ($p_T \geq 1.5$ GeV/c).

region	hits	muons	ε_∞	c	plain average (%)	
			(%)	((GeV/c) ²)	> 10 GeV/c	all
VXD barrel	53 750	13 440	97.80 ± 0.09	$+0.27 \pm 0.21$	97.89	97.82
VXD endcap	19 720	7 068	99.79 ± 0.05	-0.11 ± 0.13	99.76	99.83
IT barrel	24 920	10 887	99.91 ± 0.03	$+0.05 \pm 0.06$	99.92	99.89
IT endcap	40 022	8 248	97.54 ± 0.12	$+0.44 \pm 0.26$	97.69	97.18
OT barrel	19 670	7 382	99.93 ± 0.03	-0.11 ± 0.08	99.89	71.43
OT endcap	23 789	6 812	99.56 ± 0.07	-0.22 ± 0.16	99.48	71.89
VXD	73 470	14 629	98.33 ± 0.07	$+0.19 \pm 0.15$	98.39	98.35
IT + OT	108 401	14 974	98.97 ± 0.05	$+0.08 \pm 0.11$	99.00	87.31
ALL	181 871	14 974	98.71 ± 0.04	$+0.12 \pm 0.09$	98.75	91.74

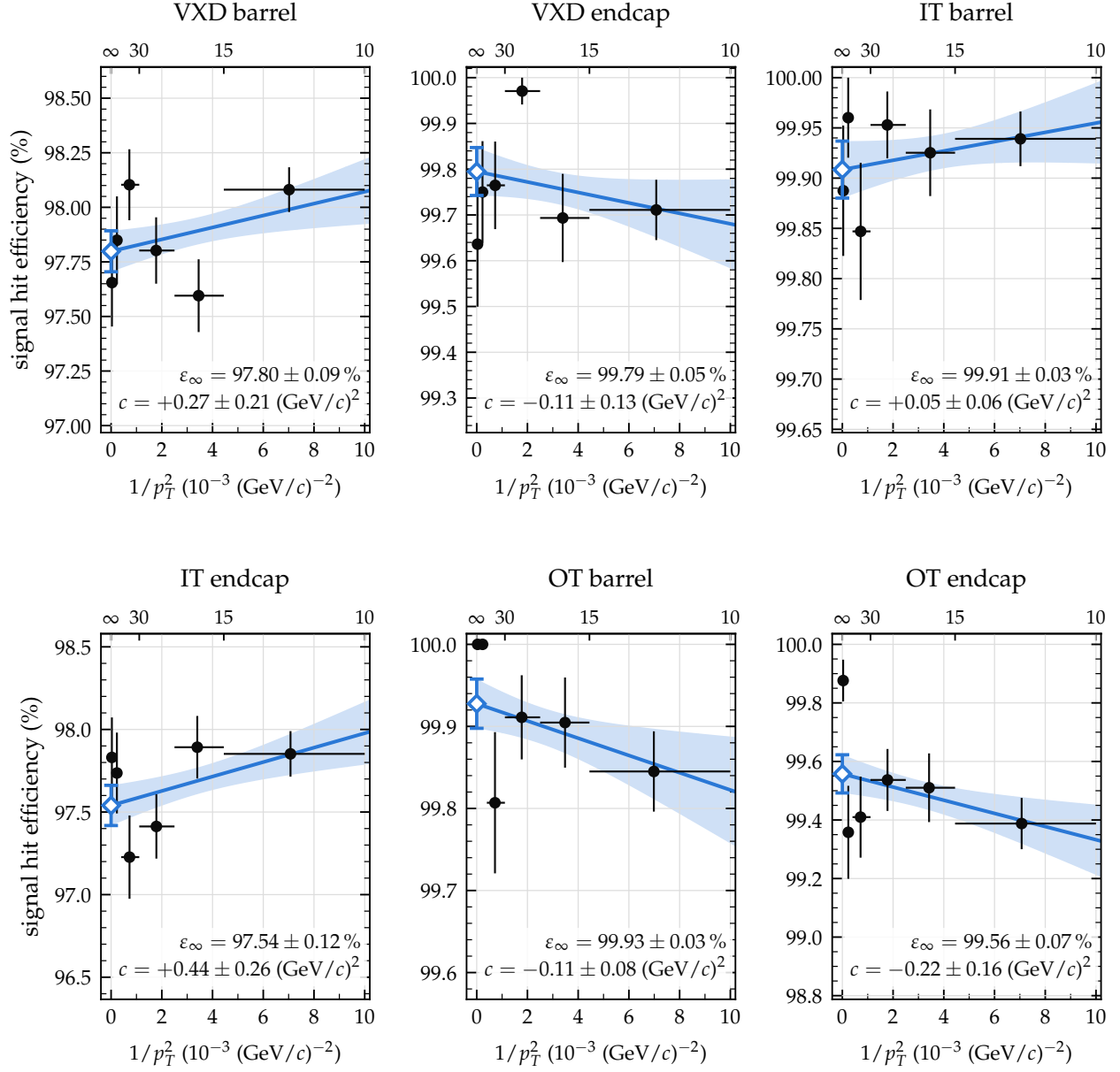


Figure 2: Signal hit efficiency against $1/p_T^2$ above $10 \text{ GeV}/c$ in the six subsystems, with the fitted line (symbols as in Fig. 1b; top axis: p_T in GeV/c). The errors of the points are computed per muon, like those of the fit. Note the different vertical scales.